Physics for Engineers 2 Lab

BUOYANCY

MPS Department | FEU Institute of Technology

OBJECTIVES

- To determine the buoyant force of an object using water displacement method.
- To determine the buoyant force of an object using the weight difference.

Buoyancy

- Buoyancy is the ability of the object to float in fluid (gas or liquid)
- Buoyant force is the upward force acted on a partially or fully submerged object. It acts opposite the gravitational force.
- In the following experiment, two ways of determining the buoyant force are presented: the fluid displacement method and the loss of mass method.

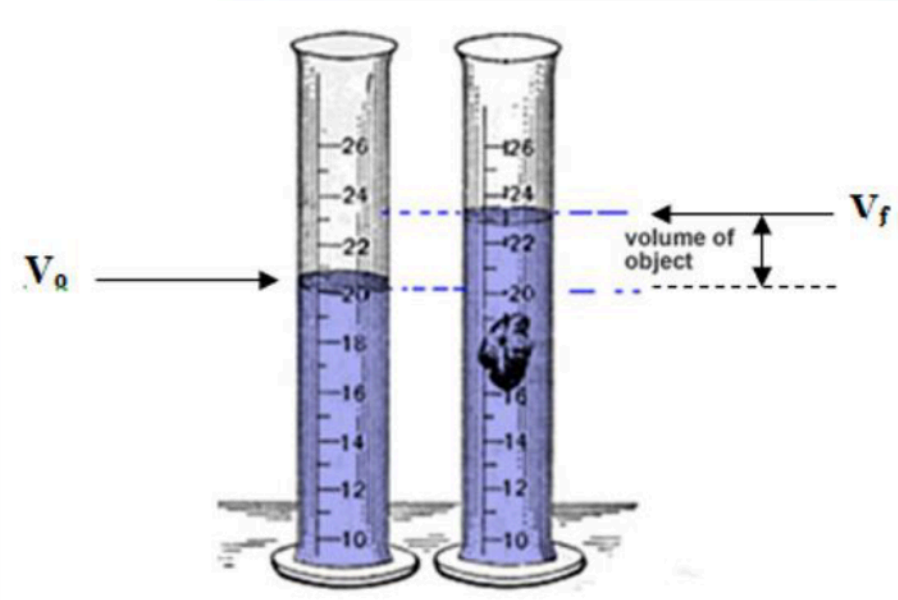
Fluid Displacement Method

In this method, the block is fully submerged in different types of fluid and the fluid is displaced. The volume of the fluid displaced is the same as the volume of the block.

$$V_{displaced} = V_{block}$$

The buoyant force experienced by the block is

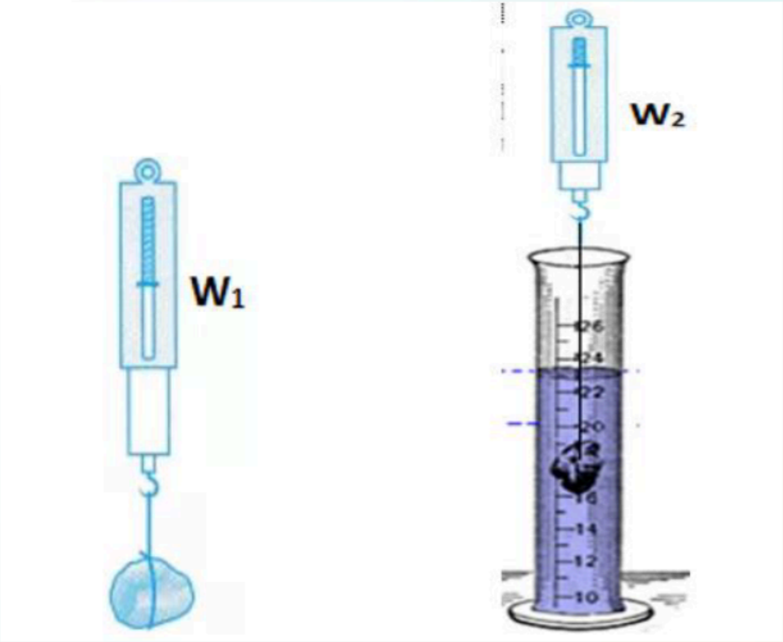
$$B_1 = W_{fluid\ displaced} = m_{fluid\ displace} \cdot g$$
$$= \rho_{medium} \cdot g \cdot V_{displaced}$$
$$B_1 = \rho_{medium} \cdot V_{displaced} \cdot g$$



where B_1 is the buoyant force
ρ_{medium} is the density of the fluid
$V_{displaced}$ is the volume of the fluid displaced
g is the gravitational acceleration
g = 9.80 m/s^2

Loss of Mass Method

In this method, the block is weighed in air and then while it is fully submerged in different types of fluid. The apparent weight of the block, that is, when it is submerged is lower than the its weight in air. This is due to buoyant force.



The buoyant force experienced by the block is

$$\boldsymbol{B}_2 = W_1 - W_2$$

where $\boldsymbol{B}_2$ is the buoyant force
W_1 is the weight of the block in air
W_2 is the weight of the block in water

Percent Difference

To get the percent difference between the buoyant force from the two methods, use the equation

$$\%\,difference = \frac{|\ \boldsymbol{B}_2 - \ \boldsymbol{B}_1|}{\frac{1}{2}(\ \boldsymbol{B}_2 + \ \boldsymbol{B}_1)} x100$$

where $\boldsymbol{B}_1$ is the buoyant force using the fluid displacement method
$\boldsymbol{B}_2$ is the buoyant force using the loss of mass method

DATA AND RESULT

The following tables are to be used in gathering data from the experiment. The data will be used in Summative Assessment.

Table 1.1 Fluid Displacement Method

QUANTITY	Medium		
	water	kerosene	brine solution
Density, ρ, (kg/m^3)	1000	820	1030
Displaced fluid, $V_{displaced}$, (m^3)			
Buoyant force, B_1 (N)			

Table 1.2 Loss of mass method

QUANTITY	Medium		
	water	kerosene	brine solution
Weight of the steel in the air, W_1 (N)			
Weight of steel in the medium, W_2 (N)			
Buoyant force, B_2 (N)			

Table 1.3 Percent Difference of Buoyant Force

	water	kerosene	brine solution
%Difference			

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